

CSC1018 - Logic

Reasoning in Predicate Logic

David Sinclair

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Gentzen System $\mathcal{G}$

The *Gentzen System* $\mathcal{G}$ for first-order predicate logic is an extension of the Gentzen System for proposition logic. It is a deductive system composed of *axioms* that are sets of formulae containing complimentary pairs of literals. The *rules of inference* are the rules for α - and β -formulae from propositional logic, plus the following rules for δ - and γ -formulae.

$$\frac{\delta \quad \delta(a)}{\frac{\forall x.A(x) \quad A(a) \quad \neg \exists x.A(x) \quad \neg A(a)}{\vdash U \cup \{\delta(a)\}} \quad \vdash U \cup \{\delta\}}$$

$$\frac{\gamma \quad \gamma(a)}{\frac{\exists x.A(x) \quad A(a) \quad \neg \forall x.A(x) \quad \neg A(a)}{\vdash U \cup \{\gamma, \gamma(a)\}} \quad \vdash U \cup \{\gamma\}}$$

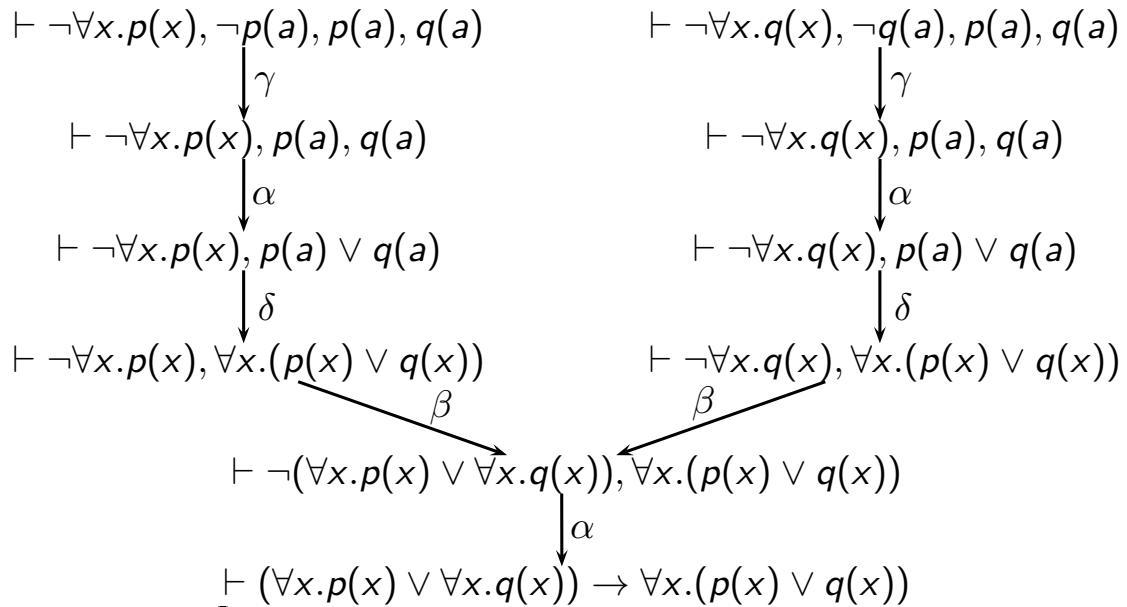
A δ -formula rule can only be applied if the constant a does not occur in any formula of U .

Gentzen System $\mathcal{G}$ Examples

Example: Using System $\mathcal{G}$ prove

$$(\forall x.p(x) \vee \forall x.q(x)) \rightarrow \forall x.(p(x) \vee q(x))$$

Technically, we should start with axioms and then apply the rules of inference to derive the target expression.



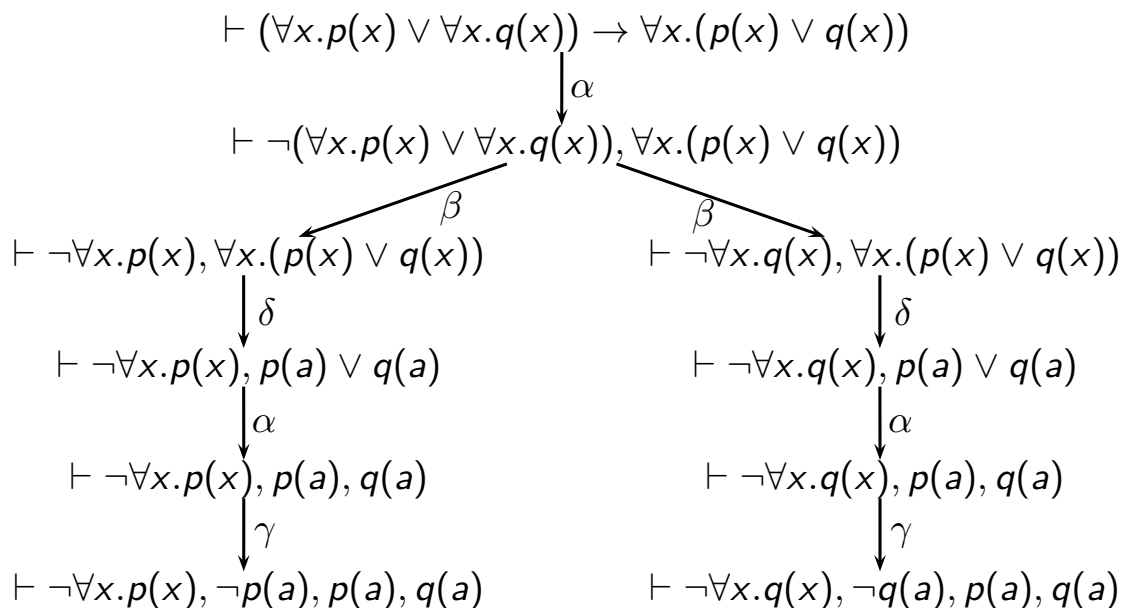
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Gentzen System $\mathcal{G}$ Examples (2)

However, we will do it in reverse!



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Gentzen System $\mathcal{G}$ Examples (3)

Example: Using System $\mathcal{G}$ prove $\exists x.\forall y.p(x, y) \rightarrow \forall y.\exists x.p(x, y)$

In reverse!

$$\begin{array}{c}
 \exists x.\forall y.p(x, y) \rightarrow \forall y.\exists x.p(x, y) \\
 \downarrow \alpha \\
 \neg\exists x.\forall y.p(x, y), \forall y.\exists x.p(x, y) \\
 \downarrow \delta \\
 \neg\forall y.p(a, y), \forall y.\exists x.p(x, y) \\
 \downarrow \delta \\
 \neg\forall y.p(a, y), \exists x.p(x, b) \\
 \downarrow \gamma \\
 \neg\forall y.p(a, y), \exists x.p(x, b), p(a, b) \\
 \downarrow \gamma \\
 \neg\forall y.p(a, y), \neg p(a, b), \exists x.p(x, b), p(a, b)
 \end{array}$$

Prenex Conjunctive Normal Form (PCNF)

The generalisation of conjunctive normal form (CNF) to first-order logic is called *prenex conjunctive normal form* (PCNF).

A formula is a *prenex conjunctive form* if and only if the formula is in the form

$$Q_1x_1 \cdots Q_nx_n M$$

where Q_i are quantifiers and M is a quantifier-free formula in conjunctive normal form. The quantifiers $Q_1x_1 \cdots Q_nx_n$ is called the *prefix*. The formula M is called the *matrix*.

Example: The following formula is in PCNF

$$\forall x.\exists y.\forall z.((q(y) \vee p(x, z)) \wedge (p(x, y) \vee q(z)) \wedge (\neg q(z)))$$

Clausal Form

A *closed formula in PCNF* has a prefix that only consists of universal quantifiers.

The *clausal form* consists of the matrix written as a set of clauses.

Example:

$$\forall x. \forall y. \forall z. ((\neg p(x) \vee q(x, y)) \wedge (p(r(x)) \vee \neg q(x, z)))$$

in clausal form is

$$\{\{\neg p(x), q(x, y)\}, \{p(r(x)), \neg q(x, z)\}\}$$

Skolem's Algorithm

In propositional logic every formula has an equivalent formula in CNF. But this is not true in first-order predicate logic.

However if A is a *closed formula* then there is a formula A' in clausal form such that A is satisfiable if and only if A' is satisfiable, ($A \approx A'$).

Consider the formula $\forall x. \exists y. p(x, y)$. This can be read as produce a value of y associated with x such that $p(x, y)$ is true.

Hence, $y = f(x)$ and we can replace the existential variable y by a function of the preceding universal variables. In this case $f(x)$ so we have $\forall x. p(x, f(x))$. This is the basic idea behind *Skolem's Algorithm* which finds the corresponding satisfiable formula in PCNF for any first-order formula.

Skolem's Algorithm (2)

Skolem's Algorithm:

We will explain the steps of the algorithm using the following example:

$$\forall x.(p(x) \rightarrow q(x)) \rightarrow (\forall x.p(x) \rightarrow \forall x.q(x))$$

1. Rename bound variables so no variable appears in 2 quantifiers.

$$\forall x.(p(x) \rightarrow q(x)) \rightarrow (\forall y.p(y) \rightarrow \forall z.q(z))$$

2. Rewrite all operators other than $\wedge$, $\vee$ and $\neg$.

$$\neg \forall x.(\neg p(x) \vee q(x)) \vee \neg \forall y.p(y) \vee \forall z.q(z)$$

3. Push negations inwards, removing double negations, using $\neg \forall x.A(x) \equiv \exists x.\neg A(x)$ and $\neg \exists x.A(x) \equiv \forall x.\neg A(x)$.

$$\exists x.(p(x) \wedge \neg q(x)) \vee \exists y.\neg p(y) \vee \forall z.q(z)$$

Skolem's Algorithm (3)

4. Extract quantifiers from the matrix, using

$$A \text{ op } Qx.B(x) \equiv Qx.(A \text{ op } B(x)) \text{ and } Qx.A(x) \text{ op } B \equiv Qx.(A(x) \text{ op } B).$$

$$\exists x.\exists y.\forall z.((p(x) \wedge \neg q(x)) \vee \neg p(y) \vee q(z))$$

5. Use the distributivity laws to transform the matrix into CNF.

$$\exists x.\exists y.\forall z.((p(x) \vee \neg p(y) \vee q(z)) \wedge (\neg q(x) \vee \neg p(y) \vee q(z)))$$

6. For every existential quantifier $\exists x$ with *preceding* universal quantifiers $\forall y_1 \dots \forall y_n$ replace x by a *new* n -ary function $f(y_1 \dots y_n)$. If there are no preceding universal quantifiers replace x by a *new* constant.

$$\forall z.((p(a) \vee \neg p(b) \vee q(z)) \wedge (\neg q(a) \vee \neg p(b) \vee q(z)))$$

7. Write the formula in clausal form dropping the universal quantifiers.

$$\{\{p(a), \neg p(b), q(z)\}, \{\neg q(a), \neg p(b), q(z)\}\}$$

Skolemisation Example

Original Formula	$\exists x.\forall y.p(x, y) \rightarrow \forall y.\exists x.p(x, y)$
Rename bound variables	$\exists x.\forall y.p(x, y) \rightarrow \forall w.\exists z.p(z, w)$
Eliminate Operators	$\neg\exists x.\forall y.p(x, y) \vee \forall w.\exists z.p(z, w)$
Push negation inwards	$\forall x.\exists y.\neg p(x, y) \vee \forall w.\exists z.p(z, w)$
Extract quantifiers	$\forall x.\exists y.\forall w.\exists z.(\neg p(x, y) \vee p(z, w))$
Distribute matrix	no change
Replace $\exists$	$\forall x.\forall w.(\neg p(x, f(x)) \vee p(g(x, w), w))$
Clausal form	$\{\neg p(x, f(x)), p(g(x, w), w)\}$

Unification

A *grounded term* is a term that contains constants and/or predicates of constants.

It is clear that the grounded terms $p(a, b)$ and $\neg p(a, b)$ are complementary literals and clauses containing them would be clashing clauses.

The literals $p(f(x), g(y))$ and $\neg p(f(f(a)), g(z))$ do not appear to be complementary unless we consider the following substitution.

$$\theta_1 = \{x \leftarrow f(a), y \leftarrow f(g(a)), z \leftarrow f(g(a))\}$$

Under the substitution θ_1 we get the terms $p(f(f(a)), g(f(g(a))))$ and $\neg p(f(f(a)), g(f(g(a))))$ which are complementary.

Unification (2)

There are simpler substitutions that cause the terms to be complementary and their containing clauses to clash.

$$\theta_2 = \{x \leftarrow f(a), y \leftarrow a, z \leftarrow a\}$$

The substitution $\mu = \{x \leftarrow f(a), z \leftarrow y\}$ is the simplest substitution that causes clashing clauses and hence results in the most general result in the resolution process.

Given a set of atoms $U = \{A_1, \dots, A_n\}$ the *most general unifier* (*mgu*), μ , is the simplest substitution such that applying the unifier μ to the atoms $A_1, \dots, A_n$ results in $A_1\mu = \dots = A_n\mu$

Unification Algorithm

Given a set of term equations apply the following transformations as long as one of them is applicable.

1. Transform $t = x$, where t is not a variable, to $x = t$.
2. Erase equations of the form $x = x$.
3. For an equation $f(t'_1, \dots, t'_k) = g(t''_1, \dots, t''_k)$
 - If $f \neq g$ terminate and report the terms are not unifiable.
 - Otherwise, replace $f(t'_1, \dots, t'_k) = g(t''_1, \dots, t''_k)$ by k equations $t'_1 = t''_1, \dots, t'_k = t''_k$
4. Let $x = t$ be an equation such that x has another occurrence in the set.
 - If x occurs in t and x differ from t terminate and report the terms are not unifiable.
 - Otherwise, transform the set by replacing all occurrences of x in other equations by t .

Unification Example

Example: Unify $g(y) = x$ and $f(x, h(x), y) = f(g(z), w, z)$.

Applying rule 1 gives $x = g(y), f(x, h(x), y) = f(g(z), w, z)$.

Applying rule 3 to the second equation gives:

$$x = g(y), x = g(z), h(x) = w, y = z$$

Applying rule 4 to the second equation and replacing x by $g(z)$ yields:

$$g(z) = g(y), x = g(z), h(g(z)) = w, y = z$$

Applying rule 3 to the first equation and yields:

$$z = y, x = g(z), h(g(z)) = w, y = z$$

Unification Example (2)

Applying rule 4 to the last equation and replacing y by z and removing identity equations, i.e. $z = z$, yields:

$$x = g(z), h(g(z)) = w, y = z$$

Applying rule 1 to the second equation yields:

$$x = g(z), w = h(g(z)), y = z$$

Since rules 1-4 cannot be applied to any of the equations, then the most general unifier (mgu) is:

$$\mu = \{x \leftarrow g(z), w \leftarrow h(g(z)), y \leftarrow z\}$$

General Resolution

If two clauses C_1 and C_2 contain literals $L_1 = \{l_1^1, \dots, l_{n_1}^1\} \subseteq C_1$ and $L_2 = \{l_1^2, \dots, l_{n_2}^2\} \subseteq C_2$ such that L_1 and L_2^c can be unified by an mgu μ , then C_1 and C_2 are clashing clauses and the resolvent is:

$$\text{Res}(C_1, C_2) = (C_1\mu - L_1\mu) \cup (C_2\mu - L_2\mu)$$

General Resolution Procedure

Given a set of clauses S , let $S_0 = S$.

- For any S_i , choose clashing clauses C_1, C_2 .
 - Let $C = \text{Res}(C_1, C_2)$.
 - If $C = \square$ then terminate and S is *unsatisfiable*.
 - Otherwise $S_{i+1} = S_i \cup C$.
- If $S_{i+1} = S_i$ the terminate and S is satisfiable.

General Resolution Example

Show that the following formula in clausal form is unsatisfiable.

$$\{\{\neg p(x), q(x), r(x, f(x))\}^1, \{\neg p(x), q(x), s(f(x))\}^2, \{t(a)\}^3, \{p(a)\}^4, \\ \{\neg r(a, y), t(y)\}^5, \{\neg t(x), \neg q(x)\}^6, \{\neg t(x), \neg s(x)\}^7\}$$

The clauses are numbered for referencing purposes.

Unify clauses 3 and 6 with $mgu = \{x \leftarrow a\}$
 $\{\neg q(a)\}^8$

Unify clauses 2 and 4 with $mgu = \{x \leftarrow a\}$
 $\{q(a), s(f(a))\}^9$

Unify clauses 8 and 9
 $\{s(f(a))\}^{10}$

Unify clauses 1 and 4 with $mgu = \{x \leftarrow a\}$
 $\{q(a), r(a, f(a))\}^{11}$

General Resolution Example (2)

Unify clauses 8 and 11

$$\{r(a, f(a))\}^{12}$$

Unify clauses 5 and 12 with $mgu = \{y \leftarrow f(a)\}$

$$\{t(f(a))\}^{13}$$

Unify clauses 7 and 13 with $mgu = \{x \leftarrow f(a)\}$

$$\{\neg s(f(a))\}^{14}$$

Unify clauses 10 and 14

$$\{\square\}^{15}$$

Hence the formula is unsatisfiable.

Resolution in Practice

Given the statements that all dogs are animal, all animal will die and that fido is a dog, how do we use resolution to prove fido will die?

Remember that a *Prolog-style* statement:

$$A :- B, C, D.$$

is equivalent to:

$$\begin{aligned} & A \leftarrow B \wedge C \wedge D \\ \equiv & \neg(B \wedge C \wedge D \wedge \neg A) \quad \text{since } X \rightarrow Y \equiv \neg(X \wedge \neg Y) \end{aligned}$$

Since Resolution is a *refutation* procedure, in order to prove $A \leftarrow B \wedge C \wedge D$, we need to refute $B \wedge C \wedge D \wedge \neg A$.

First, convert facts into predicate logic:

- $\forall x. (dog(x) \rightarrow animal(x))$ *All dogs are animals.*
- $dog(fido)$ *Fido is a dog.*
- $\forall x. (animal(x) \rightarrow die(x))$ *All animals die.*

Resolution in Practice (2)

Convert these predicate logic statements into clausal form (remembering to rename bound variable)

1. $\neg \text{dog}(x) \vee \text{animal}(x) = \{\neg \text{dog}(x), \text{animal}(x)\}$
2. $\text{dog}(\text{fido}) = \{\text{dog}(\text{fido})\}$
3. $\neg \text{animal}(y) \vee \text{die}(y) = \{\neg \text{animal}(y), \text{die}(y)\}$

Negate the conclusion:

4. $\neg \text{die}(\text{fido})$

Unify clauses 1 and 3 with $mgu = \{x \leftarrow y\}$
 $\{\neg \text{dog}(y), \text{die}(y)\}^5$

Unify clauses 2 and 5 with $mgu = \{y \leftarrow \text{fido}\}$
 $\{\text{die}(\text{fido})\}^6$

Unify clauses 4 and 6
 $\{\square\}^7$

Resolution Example 2

Dermot, Ann and Tom belong to the Outdoors Society. Every member of the Outdoors Society is either a skier or a cyclist. No cyclist likes rain and all skiers like snow. Tom dislikes whatever Dermot likes and likes whatever Dermot dislikes. Dermot likes rain and snow. Using **resolution**, determine is there a member of the Outdoors Society who is a cyclist but not a skier?

Step 1: Define predicates

- Let $S(x)$ be true if x is a skier.
- Let $C(x)$ be true if x is a cyclist.
- Let $L(x, y)$ be true if x likes y .

Resolution Example 2 (2)

Step 2: Convert text into premises.

Every member of the Outdoors Society is either a skier or a cyclist.

$$\forall x.(S(x) \vee C(x))$$

No cyclist likes rain.

$$\neg(\exists x.(C(x) \rightarrow L(x, Rain)))$$

All skier likes snow.

$$\forall x.(S(x) \rightarrow L(x, Snow))$$

Tom dislikes whatever Dermot likes and likes whatever Dermot dislikes.

$$\forall x.(L(Dermot, x) \rightarrow \neg L(Tom, x))$$

$$\forall x.(L(Tom, x) \rightarrow \neg L(Dermot, x))$$

Resolution Example 2 (3)

Dermot likes rain and snow.

$$L(Dermot, Rain)$$

$$L(Dermot, Snow)$$

Step 3: Negate conclusion/query

Query: Is there a member of the Outdoors Society who is a cyclist but not a skier?

$$\neg(\exists x.(C(x) \wedge \neg S(x)))$$

Resolution Example 2 (4)

Problem in Predicate Logic form:

1. $\forall x.(S(x) \vee C(x))$
2. $\neg(\exists x.(C(x) \rightarrow L(x, Rain)))$
3. $\forall x.(S(x) \rightarrow L(x, Snow))$
4. $\forall x.(L(Dermot, x) \rightarrow \neg L(Tom, x))$
5. $\forall x.(L(Tom, x) \rightarrow \neg L(Dermot, x))$
6. $L(Dermot, Rain)$
7. $L(Dermot, Snow)$
8. $\neg(\exists x.(C(x) \wedge \neg S(x)))$

Resolution Example 2 (5)

Step 4: Convert to CNF using Skolemisation

1. $S(x1) \vee C(x1)$
2. $C(x2)$
3. $\neg L(x2, Rain)$

Derivation of previous 2 lines

$$\begin{aligned}
 &\neg(\exists x2.(C(x2) \rightarrow L(x2, Rain))) \equiv \\
 &\neg(\exists x2.(\neg C(x2) \vee L(x2, Rain))) \equiv \\
 &\forall x2.\neg(\neg C(x2) \vee L(x2, Rain)) \equiv \\
 &\forall x2.(C(x2) \wedge \neg L(x2, Rain)) \equiv \\
 &\{\{C(x2)\}, \{\neg L(x2, Rain)\}\}
 \end{aligned}$$

4. $\neg S(x3) \vee L(x3, Snow)$
5. $\neg L(Dermot, x4) \vee \neg L(Tom, x4)$
6. $\neg L(Tom, x5) \vee \neg L(Dermot, x5)$
7. $L(Dermot, Rain)$
8. $L(Dermot, Snow)$
9. $\neg C(x6) \vee S(x6)$

Resolution Example 2 (6)

We can rewrite this as sets of terms where the terms in each set are a disjunction.

1. $\{S(x1), C(x1)\}$
2. $C(x2)$
3. $\neg L(x2, Rain)$
4. $\{\neg S(x3), L(x3, Snow)\}$
5. $\{\neg L(Dermot, x4), \neg L(Tom, x4)\}$
6. $\{\neg L(Tom, x5), \neg L(Dermot, x5)\}$
7. $\{L(Dermot, Rain)\}$
8. $\{L(Dermot, Snow)\}$
9. $\{\neg C(x6), S(x6)\}$

Resolution Example 2 (7)

Step 5: Resolve using Unification

Resolve 1 and 9 with $mgu = \{x1 \leftarrow x6\}$

$$\{S(x6)\}^{10}$$

Resolve 10 and 4 with $mgu = \{x3 \leftarrow x6\}$

$$\{L(x6, Snow)\}^{11}$$

Resolve 11 and 5 with $mgu = \{x6 \leftarrow Tom, x4 \leftarrow Snow\}$

$$\{\neg L(Dermot, Snow)\}^{12}$$

Resolve 12 with 8 with $mgu = \{\}$

$$\{\square\}$$

Hence, there is a resolution with $x6 = Tom$.